



United International University (UIU)
Dept. of Computer Science & Engineering (CSE)
MID Exam, Trimester: Summer 2025

Course Code: CSE 113/EEE 2113; Course Title: Electrical Circuits

Total Marks: 30; Duration: 1 hour 30 min

Any examinee found adopting unfair means would be expelled from the trimester/ program as per UIU disciplinary rules.

Question 1: Answer all the questions.

(8 Marks)

A circuit element is connected to a voltage source as shown in **Figure 1a**. The charge [4+4]
entering through this element is shown in **Figure 1b**. **CO1**

- Plot the current vs time graph over the interval $0 \leq t \leq 10$ s.
- Plot the power vs time graph for the same interval.

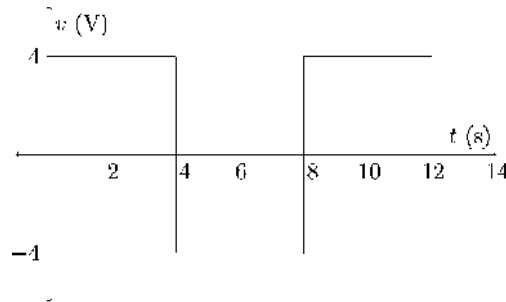


Figure 1a

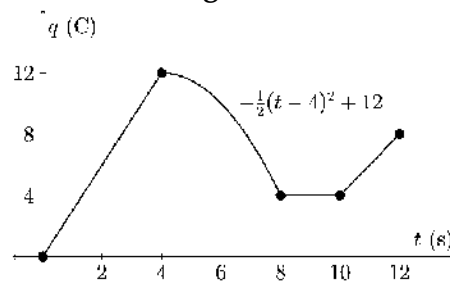


Figure 1b

Question 2: Answer all the questions.

(8 Marks)

Observe **Figure 2** and answer the followings:

- Apply KVL in Loop 1 & Loop 2 and **find** the values of I_1 and I_2
- Apply KCL at Node A & Node B and **find** the values of I_x and I_y

[4+4]

CO2

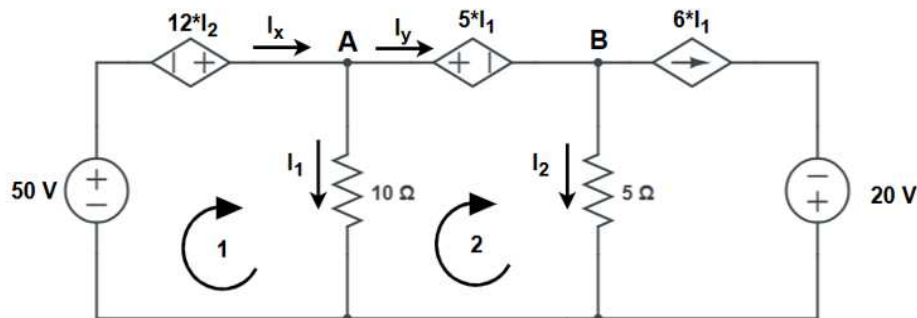
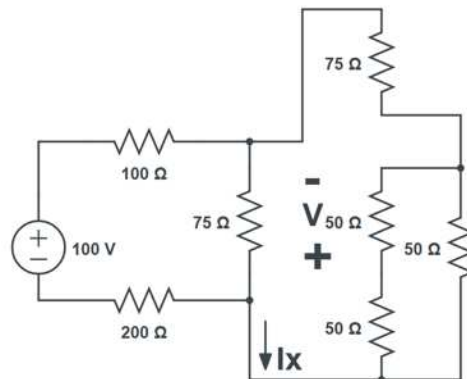


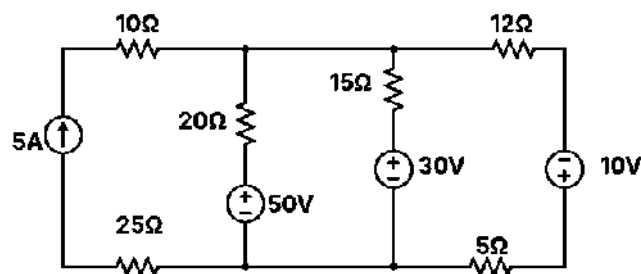
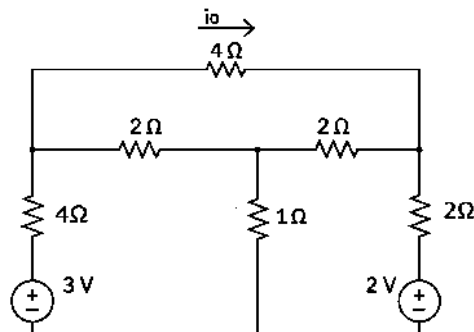
Figure 2

Question 3: Answer all the questions**(8 Marks)**Determine the following for the circuit shown in **Figure 3**:**[3+3]**

- i) Equivalent Resistance of this circuit.
- ii) Current I_x and v using CDR, VDR.

CO1**Figure 3****Question 4: Answer all the questions.****(8 Marks)**Answer the following questions for the circuit shown in **Figure 4 (a-b)**:**[4+4]**

- i) **Determine** the power of 15Ω using mesh analysis for the circuit shown in **Figure 4(a)**.
- ii) Using nodal analysis in **Figure 4b**, find the value of i_o .

CO2**Figure 4a.****Figure 4b.**